JINGTAO LEI

Applied Mathematics Undergraduate | AI & Machine Learning Research

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ACADEMIC BACKGROUND

Bachelor of Science (B.Sc.) in Applied Mathematics

Central South University, Dundee International Institute — Changsha, China
Sep 2024 – Jun 2028 (Expected) | Cumulative Average: 87.88/100

RESEARCH PROFILE

Applied Mathematics undergraduate with research experience in multimodal learning and video diffusion models. Current work at CUHK investigates whether CogVideoX-2B encodes usable physical dynamics toward world-model capabilities; earlier work developed gate-controlled prompt truncation for multimodal chest X-ray classification and led to arXiv:2609.06590.

RESEARCH INTERESTS

Multimodal learning, video diffusion, world models, representation learning, adaptive computation

RESEARCH APPOINTMENTS

Position	Organization / Department	Supervisor	Period
SURP Research Student	CUHK, Computer Science & Engineering	Prof. Shengchao Liu	Jun 2026 – Present
Undergraduate Researcher	Multimodal Learning Research Project	Prof. David P. Woodruff	Jan–Jul 2026

RESEARCH EXPERIENCE

Video Diffusion Models toward World-Model Capabilities

The Chinese University of Hong Kong — Department of Computer Science and Engineering
SURP Research Student | Supervisor: Prof. Shengchao Liu | Jun 2026 – Present

- Investigated whether pretrained video diffusion models encode usable physical dynamics, with the broader goal of extending visual generative models toward **world-model capabilities** for prediction and physical reasoning.
- Built **controlled motion and two-object interaction benchmarks** with known physical ground truth and a reproducible CogVideoX-2B analysis pipeline.
- Mapped **direction, speed, optical flow, and contact** information across Transformer layers, diffusion noise levels, and readouts using **linear probes**, **ID/OOD evaluation**, and negative controls.
- Found strong linear decodability of motion factors: best controlled ID results reached approximately 1.09° direction error and $R^2 = 0.988$ for speed; the factor map varied across layers, noise levels, and readouts.
- Designed and initiated **representation-level interventions** using factor-subspace removal, matched random orthogonal controls, and temporal perturbations to distinguish information that is merely decodable from information that is functionally used by the model.

Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier

Undergraduate Research Project

Supervisor: Prof. David P. Woodruff | *Jan 2026 – Jul 2026* | arXiv:2609.06590 | Code

- Investigated **gate-controlled prompt shortening** in parameter-efficient multimodal learning, extending a fixed-length prompt framework with a lightweight layer-wise controller.
- Extended a ViT-based **Mixture of Prompt Experts (MoPE)** classifier with **retention-ratio predictions**; mini-batch-averaged ratios determine the integer retained visual prompt length, while retained mixed prompts are scaled by individual ratios.
- Adapted and evaluated the framework on a derived IU X-Ray chest X-ray classification task with controlled **gated-versus-fixed prompting** experiments.
- In the recorded run, best validation accuracy was **0.8996 vs 0.8969** for the fixed-length baseline and final validation accuracy was **0.8963 vs 0.8802**.
- Exported gate statistics retained **one of six visual prompt tokens** at all recorded training points.

PUBLICATIONS

Jingtao Lei, Hongji Li, Dexiang Shu.
Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier.
arXiv preprint arXiv:2609.06590, 2026. | arXiv | Code

TECHNICAL SKILLS

Programming & Scientific Computing: Python, Bash, NumPy, pandas, Matplotlib

Machine Learning: PyTorch, Hugging Face Transformers & Diffusers, Vision Transformers, multimodal learning, prompt tuning, diffusion and video diffusion models

Research Methods: controlled dataset construction, representation probing, intermediate-feature analysis, ID/OOD evaluation, negative controls, ablation studies, representation-level interventions

Systems & Research Computing: Linux, SSH-based remote development, Git, Slurm, Singularity, GPU/HPC workflows; hands-on experience running experiments on NVIDIA H200 clusters

AWARDS & ACADEMIC ACHIEVEMENTS

- **Second Prize**, 6th "Celian Cup" Mathematical Modeling Elite League (Provincial-level Competition), 2026
- **First Prize**, 21st Central South University Mathematical Modeling Competition and Internal Selection for the National Competition
- **Third Prize**, Annual Comprehensive Student Evaluation Award, Central South University
- **97.80/100 (A+)**, **From Combinatorics to Calculus**, taught by Prof. Dan Ciubotaru, University of Oxford

ADDITIONAL TRAINING

AWS Academy Graduate — Cloud Foundations